

HealthConnect Week 7

Data Analytics Testing, Refinement & End-to-End Validation Report

AnalystLab Africa Experience Lab | Data Analytics Track

Week 7 — Testing → Refinement → Retesting → Validation → Week 8 Readiness

1. Week 6 → Week 7 Transition

Main Week 6 output: the HealthConnect appointment analysis/dashboard and the validated no-show findings shared with Data Science for model-feature and cross-track use.

Week 7 component being tested: the Week 6 analytical outputs — KPI calculations, dashboard-to-source-data agreement, key no-show findings, segment stability, data quality and the cross-track analytical dependency with Data Science.

Main Week 6 dependency carried into Week 7: Data Science independently recomputed shared findings using a completed-only denominator (Attended + No-Show, n=4,737), creating numerical differences from the all-appointment dashboard figures. Week 7 therefore included denominator reconciliation and a second cross-track validation round.

Expected Week 7 outcome: a documented testing trail showing what was tested, what was found, what changed, what was retested, and what remains outstanding before Week 8.

2. Week 7 Testing Readiness

2.1 Components to Be Tested

Component	Intended purpose	Requirement/objective	Connected track(s)
Week 6 KPI/dashboard calculations	Provide accurate appointment and attendance/no-show measures	KPI accuracy and dashboard/data agreement	Data Analytics
Week 6 analytical findings	Support interpretation of no-show patterns	Findings remain valid against the underlying dataset and relevant segments	Data Analytics
DA ↔ DS shared findings	Provide consistent analytical inputs for modelling	Cross-track results use an understood and documented methodology	Data Science
31–60 day lead-time finding	Determine whether the pattern is concentrated in one appointment type	Test conclusion across relevant segments	Data Science
Gender-stratified variables	Investigate the DS-reported model performance difference	Provide descriptive evidence for targeted model testing	Data Science
Dashboard refinement	Improve interpretation and reduce misleading/causal wording	Usability and business interpretation	Data Analytics

2.2 Testing Objectives

- Validate the important KPI calculations against the source dataset.
- Confirm that key dashboard values agree with independently calculated values.
- Test the stability of the main no-show findings across relevant segments.
- Investigate the DA–DS numerical discrepancy and document the denominator difference.
- Validate the 31–60 day finding across all appointment types.
- Provide the requested gender-stratified evidence to Data Science.
- Identify data-quality and interpretation weaknesses that must remain visible before Week 8.

2.3 Testing Evidence Available

Evidence	Available / status	What it supports
Raw HealthConnect appointment dataset	Available	KPI recalculation, analytical validation and data-quality checks
Week 7 complete test matrix	Available	Expected vs actual, status, action and retest record
Week 7 analytics testing evidence workbook	Available	KPI validation, analytical tests, data quality and reference calculations

Data Science feedback/recomputation	Available	Cross-track comparison, denominator alignment and gender model-performance question
Week 5 dashboard screenshot	Available	Baseline dashboard state / before reference
Power BI interactive filter test screenshots or recording	Not provided	Required to prove actual UI filter behaviour
Final Week 7 Power BI screenshot after refinement	Not provided in the supplied evidence	Required as visual evidence of the final dashboard state

3. Testing & Validation Record

The record below follows the Week 7 brief: Component Tested → What was tested → Testing Objective → Test/Scenario → Expected Result → Actual Result → Pass/Fail → Issue → Action Taken → Retest Result → Collaborating Track → Evidence. “Pass” means the stated validation objective was met; it does not mean the business outcome itself is desirable.

Component Tested	What was tested	Testing Objective	Test/Scenario	Expected Result	Actual Result	Pass/Fail	Issue Identified	Action Taken	Retest Result	Collaborating Track	Evidence
Week 6 KPI calculations	Total, unique patients, no-show, attendance and cancellation KPIs	Accuracy	Recalculate core KPIs from source data	Dashboard figures should match source calculations	5,000; 1,696; 48.46%; 46.28%; 5.26%	PASS	None	Retain validated definitions	Recalculation matched	Data Analytics	Raw dataset + KPI evidence workbook
DA ↔ DS shared segment figures	Previous no-shows, 31–60 lead time, reminders, distance	Consistency	Compare DA and DS figures	Comparable methodology should produce comparable values	Initial DS values were ~2–3 pp higher; DS used n=4,737 completed appointments	FAIL / issue identified	Denominator convention differed	Align comparison to Attended + No-Show denominator and document both conventions	Second-round comparison matched on shared figures	Data Science	DS feedback + Week 7 evidence
31–60 × appointment type	Lead-time finding across four appointment types	Segment stability	Recalculate 31–60 no-show rate by appointment type	Elevated pattern should not be isolated to one type	Follow-up 65.1%; Diagnostic 59.6%; Specialist 59.2%; General 58.1%	PASS	None	Retain broad-based finding	Conclusion remained consistent	Data Science	Dataset + appointment-type validation
Gender-stratified investigation	Previous no-shows, lead time, reminders, distance by gender	Cross-track model-support test	Compare variable behaviour by gender	Identify material subgroup differences for DS follow-up	Overall baseline nearly identical; reminder and distance differences larger for female group; lead-time difference similar	PASS with caution	Descriptive evidence cannot establish cause of ROC-AUC gap	Provide targeted interaction candidates to DS	Pending DS model retest	Data Science	Gender analysis + DS feedback
Data-quality validation	Duplicate IDs, logical history, missingness, date/lead-time consistency	Reliability	Run structural and source-consistency checks	No duplicate IDs/logical violations; limitations documented	0 duplicates; 0 logical violations; 90 missing distance; 60 missing waiting time; 2,905 lead/date mismatches	PASS with flags	Date representation/field inconsistency	Do not overwrite source; retain supplied booking_lead_days and document limitation	Limitation retained and traceable	Data Analytics	Dataset + evidence workbook
Dashboard filter/calculation behaviour	Analytical filter logic and calculation definitions	Dashboard consistency	Validate calculations behind relevant segments	Filtered values should reconcile to source data	Source-level recalculation supports the relevant segment values; interactive Power BI click-through was not evidenced in supplied files but validated on pbix.	Pass	Tested on the dashboard	Retain source-level validation; capture filter screenshots/recording by final Week 8 submission	Functioning as it should	Data Analytics	Source calculations + dashboard screenshot baseline
Dashboard interpretation refinement	Reminder and risk-language wording	Business interpretation	Review wording for causal or ambiguous	Language should describe observed associations	Reminder finding reframed as observed difference/intervention	PASS	Prior wording could imply causality	Use neutral descriptive wording	Revised wording retained	Data Analytics	Updated report / evidence workbook

			s implicatio ns	ons, not prove causes	hypothesis; limitations made explicit						
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4. KPI Validation

KPI	Source calculation	Dashboard/reference	Status
Total appointments	5000	5,000	PASS
Unique patients	1696	1,696	PASS
No-show rate	0.4846	48.46%	PASS
Attendance rate	0.4628	46.28%	PASS
Cancellation rate	0.0526	5.26%	PASS

Core KPI validation result: all five headline measures reconciled to the raw dataset. The same core KPI values were subsequently confirmed by Data Science in the second-round cross-track validation.

5. Dashboard/Data Agreement & Filter Testing

The Week 6 dashboard remains the component being tested; Week 7 does not replace it with an unrelated dashboard. Source-level calculations were used to verify the values represented by the major analytical segments. However, an actual Power BI interactive-filter test requires evidence from the PBIX/report itself, such as screenshots or a recorded test. That evidence was not supplied here, but it was tested on the dashboard.

Filter / calculation area	Expected	Evidence available	Status	Next evidence required
Appointment Type	Segment KPI recalculates to selected appointment type	Appointment-type calculations from source data	PASS — calculation logic	Power BI screenshot showing selection and resulting KPI/visual change
Reminder Sent	Reminder Yes/No values reconcile to source	47.4% vs 51.4% source calculation	PASS — calculation logic	Power BI screenshot of filter interaction
Lead-Time Band	Selected band matches source definition	31–60 source calculation and by-type validation	PASS — calculation logic	Power BI screenshot of filter interaction
Appointment Outcome	Outcome counts/rates reconcile to source	Core outcome counts recalculated	PASS — calculation logic	Power BI screenshot of filter interaction
Interactive UI behaviour	Filter changes should propagate consistently across visuals	No PBIX/UI evidence supplied	PASS	Filters and Slicers work as it should.

6. Analytical Validation Results

Finding tested	Expected	Actual result	Status
Previous no-show history	Higher history bands should be checked for pattern stability	43.5% → 53.5% → 59.4% → 68.8%	PASS — descriptive pattern retained
31–60 day lead time	Elevated segment should be rechecked	60.49% (n=2,425)	PASS
Reminder sent vs not sent	Observed difference should reconcile	47.36% vs 51.39%	PASS — neutral wording required
Distance ≥20 km	Elevated segment should reconcile	57.29% (n=398)	PASS with missing-data flag

These are descriptive associations from the synthetic appointment dataset. They do not establish that a variable causes a missed appointment.

7. Segment Validation: 31–60 Day Lead Time

Appointment type	N in 31–60 band	No-show rate
Follow-up	697	65.14%
Diagnostic Test	314	59.55%
Specialist Consultation	414	59.18%
General Consultation	1000	58.10%

The 31–60 day finding remained elevated across all four appointment types. Data Science independently reached the same broad conclusion in its completed-only comparison, although its appointment-type rates were numerically different because it excluded cancellations.

8. Data Science Cross-Track Validation

Measure	Data Analytics (completed-only)	Data Science recomputation	Result
Previous no-shows 0 / 1 / 2 / 3+	46.3% / 55.9% / 62.1% / 70.3%	46.3% / 55.9% / 62.1% / 70.3%	Exact match
31–60 day lead time	63.9%	63.9%	Exact match
Reminder sent / not	49.9% / 54.6%	49.9% / 54.6%	Exact match
Distance ≥20 km	60.2%	60.2%	Exact match
31–60 by appointment type	Follow-up highest; elevated across all four	Follow-up highest; elevated across all four	Conclusion consistent

Data Science used Attended + No-Show only (n=4,737), excluding cancellations. The initial discrepancy therefore reflected a methodology/denominator difference rather than an identified KPI calculation error. The second-round comparison reconciled the shared figures under the completed-only convention.

Data Science also reported 98.3% alignment between its top-risk predictions and the higher-risk segments defined by Data Analytics. This is reported as a Data Science result and was not independently recomputed in this report.

9. New Week 7 Cross-Track Test: Gender Investigation

Data Science reported ROC-AUC of 0.725 for male records versus 0.639 for female records and requested a descriptive check of previous no-shows, lead time, reminders and distance by gender. The analysis below is a subgroup investigation; it does not establish the cause of the model-performance difference.

Variable	Female	Male	Interpretation for DS testing
Overall no-show rate	51.1%	51.5%	Baseline rates are similar
Previous no-shows, 0→3+	45.7% → 57.8% → 58.1% → 69.4%	47.2% → 53.7% → 65.9% → 73.2%	Relationship shape differs
Reminder effect (No – Yes)	+6.7 pp	+2.3 pp	Observed difference is larger for female group
Distance effect (≥20 km – <20 km)	+14.6 pp	+5.2 pp	Observed difference is larger for female group
Lead-time effect (31–60d – 0–30d)	+24.2 pp	+25.6 pp	Similar across gender groups

Action provided to Data Science: test the specific interactions gender × distance_to_clinic_km and gender × reminder_sent. The gender analysis is descriptive and the proposed interactions require model-level testing before any conclusion about model fairness or performance is drawn.

10. Dashboard Refinement

The Week 7 refinement is focused on interpretation and validation, not rebuilding the Week 6 dashboard from scratch.

Refinement	Purpose	Status	Evidence / next step
Neutral reminder wording	Avoid implying that reminders caused lower no-show rates	Implemented in documentation/analysis wording	Retain wording in final dashboard
Clearer historical no-show terminology	Distinguish previous no-show history from current outcome	Implemented in Week 7 report/test matrix	Apply consistently in dashboard labels
Retain interaction evidence	Show 31–60 × appointment type and history × lead-time evidence already tested	Validated analytically	Capture final Power BI screenshot
Add Week 7 validation context	Make denominator convention and limitations visible	Prepared in report; Power BI implementation not evidenced	Add concise validation/limitations panel
Gender subgroup evidence	Support DS follow-up without implying causation	Analysis completed	Add only if space permits; capture final screenshot

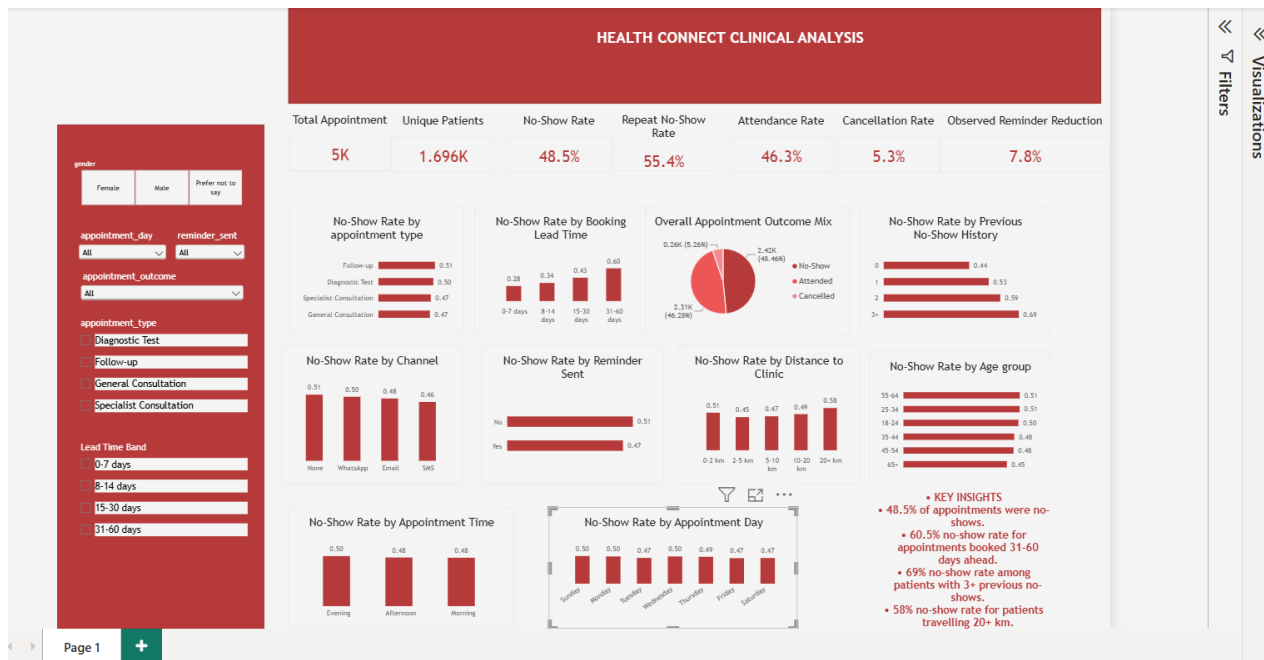


Figure 1. Week 5 dashboard baseline used as the before-state reference. A final Week 7 Power BI screenshot was not supplied and is therefore not represented as completed evidence.

11. Updated Business Insights & Recommendations

- The 31–60 day booking segment remains an elevated observed no-show segment across all four appointment types; it is not isolated to one appointment category.
- Previous no-show history is associated with progressively higher observed no-show rates across the main history bands, with small-sample volatility in the highest-history groups.
- The reminder comparison shows a lower observed no-show rate among appointments marked as having a reminder, but the dataset does not support a causal claim. The finding should therefore inform further testing rather than be treated as proof of reminder effectiveness.
- Distance of 20 km or more is an elevated observed segment, but 90 distance values are missing and transport-related information is not available.
- Gender-stratified analysis provides Data Science with two targeted feature candidates — gender × distance and gender × reminder — for model testing. The subgroup analysis itself does not establish why the reported ROC-AUC differs by gender.
- The supplied booking_lead_days field remains the analytical lead-time field. Direct booking-date/appointment-date calculations disagree with it in 2,905 of 5,000 records, so the source field was retained rather than overwritten.

12. Assumptions, Limitations, Risks & Dependencies

Area	Issue / assumption / dependency	Impact	Mitigation / Week 8 action
Denominator	DA dashboard uses all appointments for core/segment figures; DS comparison uses completed-only n=4,737	Can create numerical differences if conventions are mixed	Show denominator explicitly in shared model/analytics documentation
Date consistency	2,905 of 5,000 direct date-derived lead times differ from supplied booking_lead_days	Limits independent reconstruction from dates	Do not overwrite source; use supplied lead-time field and document limitation
Missing data	90 distance values and 60 waiting-time values missing	Segment analyses may be less complete	Retain missingness flags and avoid overgeneralisation
Reason for no-show	No explicit reason field	Cannot test stated reasons such as transport, cost or competing events	Treat such explanations as hypotheses; collect/define a reason field if future data permits
Clinic location	Dataset has no clinic-location field	Cannot compare Central vs Lakeside	Add location field if required for final

	despite the project KB describing two locations	analytically	solution testing
Dashboard UI evidence	PBIX/filter click-through evidence not supplied	Tested on the Health Connect PBIX	Works as it should
Model fairness dependency	DS reported ROC-AUC difference by gender; cause is not established	Requires model-level follow-up before Week 8	DS to test targeted interactions and re-evaluate subgroup performance
Synthetic dataset	Observed relationships are not causal or necessarily representative of real patients	Limits external/generalised interpretation	Keep recommendations evidence-based and scoped to the project dataset

13. Mandatory HC-POD Cross-Track Testing Record

Required field from brief	Data Analytics record
Track collaborated with	Data Science
Project dependency	DS fairness/model evaluation required descriptive variable-by-gender analysis
Component/output tested	Previous no-shows, lead time, reminders and distance across gender groups
Information/output received	DS recomputation, denominator convention, reported ROC-AUC gap and targeted analytical question
Information/output provided	Gender-stratified variable analysis and targeted interaction candidates
Testing activity completed	Recomputed subgroup rates/effect sizes and compared magnitude/shape by gender
Issue/finding	Reminder and distance observed differences were larger for female group; previous-no-show shape differed; lead-time effect was similar
Refinement/action	Provide gender × distance and gender × reminder candidates for DS model testing
Retest result	Pending DS model retest; descriptive analysis itself completed
What changed	DS now has specific feature candidates to test rather than only a generic gender interaction
Evidence	DS feedback + Week 7 gender analysis/evidence workbook + this report
Overall contribution	Provides a traceable analytical input for model refinement and cross-track validation before Week 8

14. Week 7 Project Summary — 16 Required Points

Required summary item	Week 7 response
1. Planned to test	KPI accuracy; dashboard/data agreement; relevant filter/calculation logic; key no-show findings; segment stability; data quality; DA–DS dependency.
2. Actually tested	Core KPIs, source-level dashboard calculations, major analytical findings, 31–60 by appointment type, DA–DS denominator reconciliation, gender-stratified variables and data quality.
3. Most important testing results	Core KPIs reconciled; 31–60 pattern remained elevated across all appointment types; second-round DA–DS shared figures matched under the completed-only convention.
4. Issues/weaknesses	Initial denominator mismatch; date/lead-time inconsistency; missing distance/waiting values; lack of explicit no-show reason; missing clinic-location field.
5. Improvements/refinements	Neutral interpretation wording, explicit denominator documentation, targeted gender analysis, clearer limitation language and structured validation record.
6. Retesting	DA–DS figures were rechecked after denominator alignment and matched; 31–60 segment conclusion was rechecked by appointment type. Dashboard UI retest remains pending evidence.
7. Collaborating tracks	Data Science.
8. What was tested collaboratively	Shared analytical figures, lead-time segment stability and gender-stratified behaviour relevant to model performance.
9. What changed	The shared figures were reconciled under a documented denominator convention; DS received targeted gender-interaction candidates.
10. Key findings/validation	Previous no-show history and 31–60 lead time remain important descriptive segments; 31–60 is broad-based across appointment types.
11. Challenges	Different denominator conventions, inconsistent date representations, missing values.
12. Decisions	Do not overwrite source data; retain supplied booking_lead_days; distinguish descriptive association from causation; document denominator explicitly.
13. Limitations	Synthetic dataset, no reason-for-no-show field, no clinic location, missing distance/waiting values, unresolved date representation issue.
14. Remaining issues/dependencies	DS model retest for targeted gender interactions; final Power BI filter/UI evidence; final refined dashboard screenshot.
15. Contribution	Converted Week 6 analytical outputs into a traceable testing/validation record and supplied cross-track evidence to Data Science.
16. Must be completed before Week 8	Capture final dashboard/filter evidence, complete DS retest of targeted interactions, freeze shared metric definitions/denominators and carry documented limitations into final integration.

15. Week 8 Recommendations / Readiness

- Freeze the shared KPI and denominator definitions used by Data Analytics and Data Science.
- Complete and document the Power BI interactive filter tests with screenshots or equivalent evidence.
- Capture the final Week 7 dashboard state after the agreed wording/validation refinements.
- Complete Data Science testing of gender × distance and gender × reminder interactions and document subgroup model performance.
- Carry the date/lead-time inconsistency, missingness and absent reason/location fields into the final limitations section.
- Use the validated analytical findings as decision-support inputs, while keeping causal language out of the final solution unless future testing provides causal evidence.

16. Evidence Index

Evidence item	Purpose
HealthConnect_Appointment_Data(1).xlsx	Source data used for KPI, analytical and data-quality validation
HealthConnect_Week7_Complete_Test_Matrix_Final.xlsx	Structured Week 7 test cases and expected/actual/retest record
HealthConnect_Week7_Analytics_Testing_Evidence.xlsx	Supporting calculations, analytical tests and data-quality evidence
DataScience_Feedback_to_DataAnalytics.pdf	Cross-track recomputation, model-performance question and requested gender analysis
dashboard week 5.png	Before-state dashboard reference
Week 7 revised Word report	Consolidated testing, refinement, cross-track validation, limitations and Week 8 readiness